MDL Midsem Master Cheat Sheet

Logic · Statistics · ML · Association Rules · Cross-Validation

1. PROPOSITIONAL LOGIC

1.1 Natural Language → Propositional Logic

Connectives:

- **and / both** ⇒ $\wedge$
- **or / either...or** ⇒ $\vee$
- **if...then / implies** ⇒ $\rightarrow$
- **not / cannot** ⇒ $\neg$
- **if and only if** ⇒ $\leftrightarrow$

Pattern table:

English	Logic
"Either A or B"	$A \vee B$
"Neither A nor B"	$\neg A \wedge \neg B$
"A only if B"	$A \rightarrow B$
"A unless B"	$\neg B \rightarrow A$
"A is necessary for B"	$B \rightarrow A$
"A is sufficient for B"	$A \rightarrow B$
"If A or B, then C"	$(A \vee B) \rightarrow C$

Key identity: $A \rightarrow B \equiv \neg A \vee B$

2024 Q1(i): "If Ram took bus (B) or drove car (D), then he arrived late (L) and missed first session (M)."
Answer: $(B \vee D) \rightarrow (L \wedge M)$ ✓

2024 Q1(ii): "If king does not castle (K) and pawn advances (P), then bishop blocked (B) or rook pinned (R)."
Answer: $(\neg K \wedge P) \rightarrow (B \vee R)$ ✓

1.2 Valid Argument Forms (MEMORISE ALL 5)

Name	Form	Abbrev
Modus Ponens	$P \rightarrow Q,\ P \vdash Q$	MP
Modus Tollens	$P \rightarrow Q,\ \neg Q \vdash \neg P$	MT
Hyp. Syllogism	$P \rightarrow Q,\ Q \rightarrow R \vdash P \rightarrow R$	HS
Disjunctive Syll.	$P \vee Q,\ \neg P \vdash Q$	DS
Resolution	$P \vee Q,\ \neg P \vee R \vdash Q \vee R$	Res

FALLACIES (NOT valid – do not pick these!):

- Affirming consequent: $P \rightarrow Q,\ Q \nvdash P$ ×
- Denying antecedent: $P \rightarrow Q,\ \neg P \nvdash \neg Q$ ×

2023 Q2 (Wizard/Muggle):
Let: A=wizard, B=muggle, C=can do magic.
Given: $A \rightarrow C$; $B \rightarrow \neg C$; B (X is muggle).
Step 1: From $B \rightarrow \neg C$ and B: **MP** ⇒ $\neg C$
Step 2: From $A \rightarrow C$ and $\neg C$: **MT** ⇒ $\neg A$
∴ X is not a wizard. **Argument is VALID.**

1.3 Truth Table Rules

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Key for $p \rightarrow q$: Only FALSE when p=T and q=F.

Tautology = all rows TRUE **Contradiction** = all rows FALSE

Essential Equivalences (memorise):

$$p \rightarrow q \equiv \neg p \vee q$$
$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$
$$p \rightarrow q \equiv \neg q \rightarrow \neg p \quad \text{(contrapositive)}$$
$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad \text{(De Morgan)}$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad \text{(De Morgan)}$$
$$(\neg p) \rightarrow (\neg q) \equiv q \rightarrow p$$

2023 Q4: $(\neg p) \Rightarrow (\neg q)$ equivalent to?
I. $p \Rightarrow q$ II. $q \Rightarrow p$
III. $(\neg q) \vee p$ IV. $(\neg p) \vee q$
Solution: $\neg p \rightarrow \neg q \equiv q \rightarrow p$ (contrapositive) $\equiv \neg q \vee p$
Answer: II and III ✓

2023 Q3 – Tautology check:
$S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$
LHS: $\neg p \wedge (p \vee q) \equiv \neg p \wedge q$
$S_1 \equiv (\neg p \wedge q) \rightarrow q \equiv p \vee \neg q \vee q \equiv T$
S_1 is a Tautology! ✓
$S_2 : q \rightarrow (\neg p \wedge (p \vee q))$
p=T, q=T: $T \rightarrow (F \wedge T) = T \rightarrow F = F$
S_2 is NOT a Tautology. ×

1.4 Resolution Proof (Proof by Contradiction)

Algorithm to prove $\Gamma \vdash \alpha$:

1. Negate the conclusion: add $\neg\alpha$
2. Convert **everything** to CNF: $A \rightarrow B$ becomes $\neg A \vee B$
3. Number each clause (C1, C2, ...)
4. Apply resolution: from $A \vee x$ and $\neg x \vee B$, derive $A \vee B$
5. Derive □ (empty clause) ⇒ proved!

2024 Q3: Prove $P \rightarrow \neg Q,\ \neg Q \rightarrow R \vdash P \rightarrow R$
Negate conclusion: $\neg(P \rightarrow R) \equiv P \wedge \neg R$ gives C3: P and C4: $\neg R$.

Clauses	Resolved	New Clause	No.
C1: $\neg P \vee \neg Q$			
C2: $Q \vee R$			
C1, C2	Q	$\neg P \vee R$	C5
C3, C5	P	R	C6
C4, C6	R	□	C7

□ derived ⇒ **Proved!** ✓

2023/2024 Q3 pattern – general template:
Prove $A \rightarrow \neg B,\ \neg B \rightarrow C \vdash A \rightarrow C$
C1: $\neg A \vee \neg B$; C2: $B \vee C$; C3: A; C4: $\neg C$
Resolve C1,C2 on B: $\neg A \vee C$ (C5).
Resolve C3,C5 on A: C (C6).
Resolve C4,C6 on C: □ ✓

2024 Q2 – Prove Resolution Rule via Truth Table:
Prove: $P \lor Q,\ \neg Q \lor R \ \Rightarrow\ P \lor R$
Build truth table for $(P \lor Q) \land (\neg Q \lor R) \to (P \lor R)$.
Show all rows = T.
Key row: P=F, Q=T, R=T: LHS=$T \land T = T$, RHS=T ✓

2. STATISTICS & NOISE

2.1 Mean and Variance

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{(mean)}$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2 \qquad \text{(population variance)}$$

Adding noise $\varepsilon \sim \mathcal{N}(m_\varepsilon, \sigma_\varepsilon^2)$:

- $\mu_{\text{new}} = \mu + m_\varepsilon$
- $\sigma^2_{\text{new}} = \sigma^2 + \sigma^2_\varepsilon$
- If $m_\varepsilon = 0$: mean **unchanged**, variance adds up

2023 Q7a: DT $= \langle 7, 8, 6, 10, 9\rangle$
$\mu = (7 + 8 + 6 + 10 + 9)/5 = 40/5 = 8$
$\sigma^2 = (1 + 0 + 4 + 4 + 1)/5 = 10/5 = 2$

2023 Q7b: Add $\mathcal{N}(0, 2)$, peak ≤ 9.
DT$' = \langle 5, 7, 6, 12, 10\rangle$, noise$= \langle -2, -1, 0, +2, +1\rangle$
New mean $= 40/5 = 8$ (unchanged, $E[\varepsilon] = 0$) ✓
Theoretical new var $= 2 + 2 = 4$.
Justify: $E[\varepsilon] = 0 \Rightarrow$ mean unchanged.
$\text{Var}(DT') = \text{Var}(DT) + \text{Var}(\varepsilon) = 2 + 2 = 4$.

3. RSS, MSE & BIAS-VARIANCE

3.1 RSS and MSE

$$\text{RSS} = \sum_{i=1}^{N}\big(\hat{f}(x_i) - y_i\big)^2$$
$$\text{MSE} = \text{RSS}/N$$

2024 Q5a: Data: $(1, 0.9), (2, 2.2), (3, 3), (4, 4.1), (5, 4.8), (6, 6)$.
Model: $\hat{f}(x) = x + 0.01x^2$ ($\theta_0 = 0, \theta_1 = 1, \theta_2 = 0.01$).
$\hat{f}$: 1.01, 2.04, 3.09, 4.16, 5.25, 6.36
RSS $= 0.0121 + 0.0256 + 0.0081 + 0.0036 + 0.2025 + 0.1296 \approx 0.383$

3.2 Bias-Variance Decomposition (FULL DERIVATION)

Setup: $y = f(x) + \varepsilon$, $E[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$, $\mu_{\hat{f}} = E[\hat{f}]$.

$$E[(y - \hat{f})^2] = E[((f - \hat{f}) + \varepsilon)^2]$$
$$= E[(f - \hat{f})^2] + \cancelto{0}{2E[(f - \hat{f})\varepsilon]} + \sigma^2$$

(ε independent of $\hat{f}$ and $E[\varepsilon] = 0$, so cross term = 0.)
Expand $E[(f - \hat{f})^2]$ by adding/subtracting $\mu_{\hat{f}}$:

$$= (f - \mu_{\hat{f}})^2 + E[(\hat{f} - \mu_{\hat{f}})^2] + \cancelto{0}{2(f - \mu_{\hat{f}})E[\mu_{\hat{f}} - \hat{f}]}$$

$$\boxed{E[(y - \hat{f})^2] = \text{Bias}^2[\hat{f}] + \text{Var}[\hat{f}] + \sigma^2}$$

where: $\text{Bias}[\hat{f}] = f(x) - E[\hat{f}]$,
$\text{Var}[\hat{f}] = E[(\hat{f} - E[\hat{f}])^2]$, $\sigma^2 = \text{Var}(\varepsilon)$

If $E[\varepsilon] = m_n \neq 0$ (2024 Q4): Cross term $\neq 0$.
Result: Bias $= f(x) + m_n - E[\hat{f}(x)]$. (mean shifts bias)

4. REGULARIZATION

4.1 Ridge

Ridge (L2): $\text{RSS} + \alpha\sum_j \theta_j^2$
– shrinks all, never exactly 0.

Lasso (L1): $\text{RSS} + \alpha\sum_j |\theta_j|$
– can drive some coefs exactly to 0.

Which is better? $\theta_2 = 0.01$:
Ridge penalty: $0.01^2 = 0.0001$ (tiny!),
Lasso penalty: $|0.01| = 0.01$ (larger).
$\Rightarrow$ **Lasso better for small noisy coefs.**

2024 Q5b: Reduce effect of x^2: add $\alpha\theta_2^2$ to RSS.

2024 Q5c(i): Weights $w_0, \ldots, w_9$, domain needs order 6.
Penalise: $\text{RSS} + \alpha(w_7^2 + w_8^2 + w_9^2)$ to drive them to 0.

2024 Q5c(ii): Domain needs order 12 but model is order 9.
Regularization cannot add terms – must rebuild model with x^{10}, x^{11}, x^{12}.

4.2 Early Stopping

Stop training when **validation error is minimum**. Training error always falls; validation error falls then rises (overfitting).

2023 Q10: Bias error $= 4/X$. Validation error $= |X - 7| + 4$.
Val. error is min when $|X - 7| = 0 \Rightarrow X = 7$. Min val error $= 4$.
Stop at iteration 7.
At $X = 10$: val error $= 3 + 4 = 7$ (vs 4 at $X = 7$).
Stopping at 7 gives $\approx 2\times$ less validation error. ✓

5. CROSS-VALIDATION

5.1 Nested Cross-Validation

Structure:

- **Outer loop:** Each fold takes a turn as the **test** set
- **Inner loop:** From remaining folds, each takes a turn as **validation**; rest = training
- Train one model per hyperparameter combo in each inner split

Counting (independent hyperparams):

$$\text{Total models} = k \times (k-1) \times |\text{HP combos}|$$

Model name: $M_{\text{train folds}}(x_i)(y_j)$
e.g., $M_{123}(x_1)(y_1)$ = trained on folds 1,2,3 with $X = x_1, Y = y_1$

2023 Q8a: 5 folds, $X \in \{x_1, x_2, x_3\}$, $Y \in \{y_1, y_2\}$ (independent).
Combos $= 3 \times 2 = 6$. Total: $5 \times 4 \times 6 = \mathbf{120}$ models.

2023 Q8b: $Y = y_1 \Rightarrow X \in \{x_1, x_2\}$; $Y = y_2 \Rightarrow X = x_3$ (dependent).
Valid combos: $(x_1, y_1), (x_2, y_1), (x_3, y_2) = 3$.
Total: $5 \times 4 \times 3 = \mathbf{60}$ models.

6. ASSOCIATION RULES

6.1 Definitions

Let N=total transactions, $n(X)$=transactions containing X.

$$\text{Support}(X) = \frac{n(X)}{N}$$

$$\text{Confidence}(X \to Y) = \frac{n(X \cup Y)}{n(X)}$$

$$\text{Lift}(X \to Y) = \frac{\text{Conf}(X \to Y)}{\text{Supp}(Y)} = \frac{n(X \cup Y) \cdot N}{n(X) \cdot n(Y)}$$

Proof: Lift$(X \to Y)$ = Lift$(Y \to X)$ (2023 Q5)

Via probabilities:

$$\text{Lift}(X \to Y) = \frac{P(X \cup Y)}{P(X) \cdot P(Y)} = \frac{P(Y \cup X)}{P(Y) \cdot P(X)} = \text{Lift}(Y \to X) \checkmark$$

Via counts:

$$\text{Lift}(X \to Y) = \frac{n(X \cup Y) \cdot N}{n(X) \cdot n(Y)} = \frac{n(Y \cup X) \cdot N}{n(Y) \cdot n(X)} = \text{Lift}(Y \to X) \checkmark$$

$X \cup Y = Y \cup X$ (sets commute) $\Rightarrow$ symmetric.

6.2 Apriori Algorithm

Anti-monotonicity: If itemset S is infrequent, ALL supersets of S are infrequent – prune them!
Steps:

1. Count all single items; prune below threshold $\to$ frequent 1-itemsets
2. Generate pairs from frequent 1-itemsets; count; prune
3. Iteration k: join $(k-1)$-itemsets; prune candidates with any infrequent $(k-1)$-subset; count; prune
4. Stop when no new frequent itemsets

2023 Q6: Threshold = 2.
T1:{1,2,3}, T2:{2,3,4}, T3:{4,5}
T4:{1,2,4}, T5:{1,2,3,5}, T6:{1,2,3,4}

Iter 1: 1:4, 2:5, 3:4, 4:4, 5:2. All ≥ 2.

Iter 2 pruned: $\{1,5\}, \{2,5\}, \{3,5\}, \{4,5\}$ (count 1).
Frequent: $\{1,2\}$:4, $\{1,3\}$:3, $\{1,4\}$:2, $\{2,3\}$:4, $\{2,4\}$:3, $\{3,4\}$:2

Iter 3: $\{1,3,4\}$:1×pruned.
Frequent: $\{1,2,3\}$:3, $\{1,2,4\}$:2, $\{2,3,4\}$:2
Iter 4: $\{1,2,3,4\}$: $\{1,3,4\}$ infrequent $\Rightarrow$ pruned. **Done.**

7. FEATURE ENGINEERING & ENCODING

7.1 One-Hot Encoding

For categorical feature with k values, create $k-1$ binary indicator columns (drop one as baseline).

Loc	Sz	L_G	S_M	S_S	Price
G	B	1	0	0	400
G	M	1	1	0	300
G	S	1	0	1	200
B	B	0	0	0	400
B	M	0	1	0	250
B	S	0	0	1	150

Baseline = Bad location, Big size.

7.2 Linear Model with Interaction Terms (2023 Q9)

Model: $\hat{P} = \theta_0 + \theta_1 L_G + \theta_2 S_M + \theta_3 S_S + \theta_4 L_G S_M + \theta_5 L_G S_S$

Solving from data:

- $(B, B) \Rightarrow \theta_0 = 400$
- $(B, M) \Rightarrow 400 + \theta_2 = 250 \Rightarrow \theta_2 = -150$
- $(B, S) \Rightarrow 400 + \theta_3 = 150 \Rightarrow \theta_3 = -250$
- $(G, B) \Rightarrow 400 + \theta_1 = 400 \Rightarrow \theta_1 = 0$
- $(G, M) \Rightarrow 400 - 150 + \theta_4 = 300 \Rightarrow \theta_4 = 50$
- $(G, S) \Rightarrow 400 - 250 + \theta_5 = 200 \Rightarrow \theta_5 = 50$

Final: $\hat{P} = 400 - 150 S_M - 250 S_S + 50 L_G S_M + 50 L_G S_S$

7.3 Feature Engineering Techniques

Binning: Continuous $\to$ discrete bins.
Ex: Age$\to$Young/Adult/Senior.

Power Transformation: Apply x^p to change distribution shape.
Right skew $\to \log(x)$ or $\sqrt{x}$. Left skew $\to x^2$.

Dimensionality Reduction: Reduce features to prevent overfitting.
PCA, feature selection, etc.

8. KEY ML CONCEPTS

8.1 Supervised vs Unsupervised

Supervised: Each example (x_i, y_i) has a label. Learn $f : X \to Y$.
Ex: spam detection, price prediction.

Unsupervised: No labels. Find structure in $x_1, \ldots, x_n$.
Ex: clustering, PCA.

8.2 AlphaGo (2023 Q6

What: ML system to play Go/chess.
Key techniques (mention all 3+):

- **Deep Neural Networks**: policy net (which move?) + value net (who wins?)
- **Monte Carlo Tree Search (MCTS)**: explores game tree
- **Reinforcement Learning**: trained by self-play
- **Supervised pre-training**: on human expert games

Result: Beat world Go champion in 2016; beat chess robot champion in 4h.

8.3 AlphaFold (2024 Q5b)

Problem: Predict 3D protein structure from amino acid sequence.
Why important: Shape = function; critical for drug discovery.
Why hard: Levinthal's paradox – astronomically many conformations.
Technology:

- Transformer-based deep learning
- Attention mechanisms (cross-domain from NLP)
- Multiple Sequence Alignments (MSA) as features

Result: Solved 50-year-old grand challenge (2020).

8.4 Overfitting and Bias-Variance Tradeoff

	Train error	Val error
Underfitting	High	High
Good fit	Low	Low
Overfitting	Very Low	High

High complexity → Low bias, High variance.
Fixes: L1/L2 regularization, more data, dim. reduction, early stopping.

QUICK FORMULA REFERENCE

Association Rules:

$$\text{Supp}(X) = n(X)/N$$
$$\text{Conf}(X \to Y) = n(X \cup Y)/n(X)$$
$$\text{Lift}(X \to Y) = \frac{n(X \cup Y) \cdot N}{n(X) \cdot n(Y)}$$

Lift is **symmetric**: $L(X \to Y) = L(Y \to X)$

Statistics:

$$\mu = \frac{1}{n} \sum x_i$$
$$\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$$

After $+\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$:
$\mu_{\text{new}} = \mu$
$\sigma^2_{\text{new}} = \sigma^2 + \sigma^2_\varepsilon$

Bias-Variance:

$$E[(y - \hat{f})^2] = \text{Bias}^2 + \text{Var} + \sigma^2$$

$\text{Bias} = E[\hat{f}] - f(x)$
$\text{Var} = E[(\hat{f} - E[\hat{f}])^2]$
σ^2= noise var (irreducible)

Logical Equivalences:

$$p \to q \equiv \neg p \vee q$$
$$\neg(p \to q) \equiv p \wedge \neg q$$
$$p \to q \equiv \neg q \to \neg p \quad \text{(contrapositive)}$$
$$(\neg p) \to (\neg q) \equiv q \to p$$
$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad \text{(De Morgan)}$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad \text{(De Morgan)}$$

Regularization:
Ridge (L2): $\text{RSS} + \alpha \sum \theta_j^2$
Lasso (L1): $\text{RSS} + \alpha \sum |\theta_j|$ (can zero out)
Custom: $\text{RSS} + \alpha \cdot$ (penalty on specific θ_j)

Nested CV: $k \times (k-1) \times |\text{HP combos}|$

Early stopping: stop at $\arg\min_x$(val error)

You know this material. Read carefully, show all steps, write justifications. Good luck!